Unified Cognition from a Single Optimization Principle: Active Inference in MeTTa with Emergent Reasoning, Planning, and Self-Knowledge

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Abstract. We present a cognitive architecture for artificial general intelligence in which perception, action selection, causal reasoning, temporal planning, and self-knowledge all derive from a single optimization principle: minimize Expected Free Energy (EFE). Implemented in approximately 12,000 lines of MeTTa, the architecture unifies three forms of inference—induction, deduction, and abduction—as hypothesis generators under a uniform metabolic selection mechanism: hypotheses that predict correctly earn energy; those that do not are removed. This yields emergent behaviors including spontaneous investigation of hidden causes (the *Sherlock Holmes effect*), principled retreat under viability threat, and honest uncertainty grounded in actual query failure. An adaptive beam search reduces planning complexity from $O(|A|^d)$ to $O(|A| \times k \times d)$, achieving an 80,000× reduction at 50 actions. A magnitude-banded covariance pruning scheme reduces pairwise Hebbian scaling from $O(E^2)$ to $O(E \cdot b)$, achieving 19.4× pair reduction at 1,000 observables with zero significant false negatives. An integrated 14-observable reef monitoring evaluation demonstrates emergent causal discovery, metabolic pruning, and—via ablation—that abductive inference is necessary for investigation behavior.

Keywords: Active Inference · Expected Free Energy · Cognitive Architecture · MeTTa · Abductive Reasoning · Structure Learning · Metabolic Selection

1 Introduction

Active Inference [6,17] proposes that adaptive agents can be understood as minimizing a single quantity—the Expected Free Energy (EFE)—which unifies perception, action, learning, and exploration under one variational principle. Yet most implementations either remain at the level of mathematical formalism, rely on neural network approximations that sacrifice transparency [15,3], or are restricted to narrow domains with hand-crafted generative models.

We present Project Dagaz, a cognitive architecture that takes Active Inference seriously as a computational principle. We use *artificial general intelligence* (AGI) in the sense defined by Goertzel and Pennachin [9]: a system capable of handling cognitive tasks across domains it was not specifically engineered for, transferring learned structure between contexts, and acquiring novel representations from experience. The system is implemented in MeTTa [19], a symbolic meta-language for the OpenCog Hyperon framework [10], and validated at three levels: executable Python specifications encoding identical dynamics, a pure Python MeTTa evaluator that loads and executes the canonical source through generic pattern matching (implementing only universal language primitives, no domain-specific logic), and an end-to-end orchestrator with an optional language model for conversational applications (Section 4). Every inference is traceable, every belief is queryable, and every action selection decision can be decomposed into its contributing terms.

The core claim is that a single optimization principle—minimize EFE—combined with a uniform metabolic selection mechanism over symbolically generated hypotheses, produces a rich set of cognitive behaviors without scripted rules or mode switches. Specifically, we demonstrate:

1. **Unified action selection.** Exploration, exploitation, and self-preservation emerge from the same EFE equation, with no thresholds or mode flags.
2. **Emergent causal reasoning.** Induction, deduction, and abduction operate as hypothesis generators feeding a single metabolic selection mechanism.
3. **The Sherlock Holmes effect.** Abduction hypothesizes hidden causes, creating low-precision beliefs whose high information gain drives spontaneous investigation—no module was programmed to produce this behavior.
4. **Efficient temporal planning.** An adaptive beam search with emergent beam width achieves up to 80,000× reduction in evaluations at 50 actions.
5. **Scalable causal discovery.** Magnitude-banded covariance pruning reduces pairwise Hebbian cost from $O(E^2)$ to $O(E \cdot b)$ with zero significant false negatives at 1,000 observables.

2 Background

2.1 Active Inference and Expected Free Energy

Active Inference [6,17] casts perception and action as a single process of free energy minimization. An agent maintains a generative model and acts to minimize prediction error. The EFE of a policy π decomposes into pragmatic and epistemic components [16]:

$$G(\pi) = \underbrace{E_\pi[\text{prediction error}] + \text{cost}(\pi)}_{\text{pragmatic value}} - \underbrace{\tfrac{1}{2}\Delta\text{precision}(\pi)}_{\text{epistemic value}} + \underbrace{V(\pi)}_{\text{viability}}, \qquad (1)$$

where the agent selects $\pi^* = \arg\min_\pi G(\pi)$. When errors are small and the agent is safe, information gain dominates and the agent explores. When viability is threatened, $V(\pi)$ dominates and the agent retreats. No mode switches are required.

2.2 Peircean Reasoning

Peirce identified three fundamental inference forms [18]: *Induction*—from observed patterns, infer a general rule; *Deduction*—from known rules, infer specific consequences; *Abduction*—from observed effects and known rules, infer the best explanation. Our architecture implements all three as hypothesis generators feeding a single metabolic selection mechanism—the same invariant governs which hypotheses survive.

2.3 MeTTa and Hyperon

MeTTa [19] is a symbolic programming language for the OpenCog Hyperon cognitive framework [10]. Programs are subgraphs of an Atomspace—a typed metagraph serving as program, data, and knowledge representation. We chose MeTTa because its unification engine natively supports the constraint solving required for causal discovery (e.g., finding transitive closures across a causal graph), its homoiconicity allows learned rules to be immediately executable (e.g., chunked action sequences injected as new EFE-evaluable policies), and its metagraph provides metabolic memory management where atom survival is governed by predictive accuracy rather than lexical scope.

3 Architecture

The architecture comprises 22 MeTTa modules totaling approximately 12,000 lines. Figure 1 provides an overview. We describe each major component below.

3.1 The EFE Invariant

The central computation is EFE minimization over candidate actions. For each action a and observable o_i with belief (v_i, p_i) (value, precision), preference $(\hat{v}_i, w_i)$ (preferred value, importance), and action model $(\delta\mu_i, \delta p_i)$ (expected change to mean and precision):

$$\mathrm{EFE}(a) = \sum_i \underbrace{|v_i + \delta\mu_i - \hat{v}_i| \cdot (p_i + \delta p_i) \cdot w_i}_{\text{expected error}} + \mathrm{cost}(a) - w_e \cdot \mathrm{IG}(a) + V(a), \quad (2)$$

where $\mathrm{IG}(a) = \mathrm{IG}_{\mathrm{obs}}(a) + \mathrm{IG}_{\mathrm{model}}(a)$ decomposes information gain into observational (precision increase from sensing) and model-based (learning about action effects) components. The epistemic weight $w_e = (1 - \text{viability pressure}) \cdot (1 - \text{precision})$, floored at 0.05, modulates exploration based on safety and uncertainty. $V(a)$ provides a viability bonus for actions moving away from existential boundaries and a penalty for those approaching them. The selected action is $a^* = \arg\min_a \mathrm{EFE}(a)$.

The MeTTa implementation directly encodes Eq. (2):

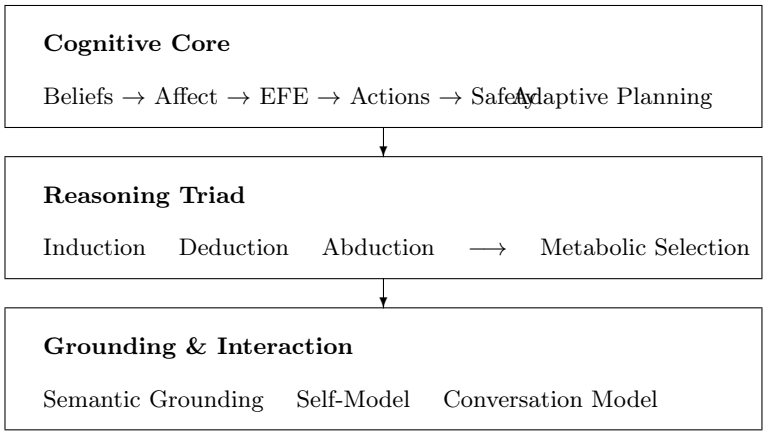

Fig. 1. Architecture overview. The cognitive core (top) runs the perception–belief–affect–EFE–action loop. The reasoning triad (middle) generates hypotheses via induction, deduction, and abduction, all subject to metabolic selection. Grounding and interaction (bottom) connect learned structure to semantic primitives and dialogue.

```
(= (compute-efe $action)
   (let* (
     ($expected-error (predict-total-error $action))
     ($cost          (get-action-cost $action))
     ($obs-ig        (compute-obs-info-gain $action))
     ($model-ig      (compute-model-info-gain $action))
     ($total-ig      (+ $obs-ig $model-ig))
     ($ew            (epistemic-weight))
     ($v-effect      (viability-effect $action))
     ($efe (+ (- (+ $expected-error $cost)
                (* $ew $total-ig))
              $v-effect))
   ) $efe))

(= (select-action-myopic)
   (let* (
     ($efes (compute-efe-list (base-actions)))
     ($best (find-min-efe-dynamic $efes))
   ) $best))
```

Three departures from the standard Active Inference formulation are noted. First, the expected error uses precision-weighted absolute deviation rather than KL divergence—a computationally tractable symbolic proxy that preserves the essential property that actions reducing expected error under high precision are preferred, while providing a smoother optimization landscape during early calibration when errors are large. Under Gaussian assumptions, the expected absolute deviation $E[|x - \mu|] = \sigma\sqrt{2/\pi}$ is a monotonic function of variance, so minimizing precision-weighted absolute error preserves the same action ordering as minimizing KL divergence when the generative model is unimodal—the regime in which Dagaz operates. Second, viability bounds are separated into an explicit $V(a)$ term rather than encoded as extreme priors, reflecting a principled dis-

tinction between survival constraints (constitutionally immutable; Section 3.6) and task preferences (modifiable). Third, the standard $\frac{1}{2}\delta p$ information gain is replaced by an adaptive epistemic weight w_e applied to decomposed information gain, derived from safety margin and residual uncertainty rather than a fixed coefficient.

Crucially, no action receives special treatment. Domain actions (`observe`, `retreat`) and conversational actions (`clarify`, `elaborate`) compete in the same EFE landscape using identical computations.

3.2 The Cognitive Cycle

Each cycle executes: (1) safety check; (2) action learning (compare predicted vs. actual effects; update action models); (3) prediction errors for each observable; (4) surprise computation $S_i = \frac{1}{2}p_i e_i^2$ with sliding-window traces; (5) affect derivation (valence, arousal, dominance from the error landscape—computed fresh, never stored); (6) structure learning as a *reflex*: suspicion update, causal link creation, deductive extension, hub detection, metabolic pruning—all fire when conditions are met, not as selected actions; (7) EFE-driven action selection; (8) Bayesian belief update; (9) structural reward for correct predictions; (10) action execution (gated by safety); (11) constitutional watchdog verification.

Structure learning is a reflex, not an action. This prevents the system from having to *choose* to learn. Generation is reflexive; retention is metabolic.

3.3 The Peircean Reasoning Triad

The architecture implements all three Peircean inference forms as hypothesis generators feeding a single selection mechanism.

Induction: Causal Discovery from Co-Surprise. When two observables exhibit correlated prediction errors across time, the system hypothesizes a causal relationship via a Hebbian accumulator in squared surprise space:

$$\Delta s_{AB}^{(\ell)} = \eta \cdot \text{sign}(e_A) \cdot \text{sign}(e_B) \cdot S_A \cdot S_B, \tag{3}$$

where $s_{AB}^{(\ell)}$ is suspicion strength at lag ℓ, η is plasticity, e_i is signed prediction error, and $S_i = \frac{1}{2}p_i e_i^2$ is surprise. Suspicion decays multiplicatively each cycle by stability factor α. When $s_{AB}^{(\ell)}$ exceeds a cost threshold, a passive model atom is created. Passive models participate in prediction (how the world evolves) but never in EFE action selection (what to do), ensuring discovered dynamics inform but do not corrupt decision-making.

Scaling via Magnitude-Banded Pruning. The Hebbian accumulator forms all ordered pairs of salient error traces each cycle—$O(E^2)$ where E is the number of active traces. At 1,000 observables this generates millions of candidates. However, most pairs have negligible covariance $S_A \cdot S_B$ and can never accumulate sufficient suspicion for promotion. From the steady-state formula

$s_\infty = \eta \cdot \mathrm{cov}/(1-\alpha)$ and the promotion cost c_ℓ, the minimum per-cycle covariance needed is:

$$\mathrm{cov}_{\min} = \frac{c_\ell \cdot (1-\alpha)}{\eta \cdot f_s}, \tag{4}$$

where f_s is a safety factor. This threshold is derived entirely from metabolic parameters—no new tuning knobs are introduced.

Each salient trace is assigned to a magnitude band $b(S) = \lfloor \log_2(S/S_{\mathrm{floor}}) \rfloor$, and pairs are formed only where $b(S_A) + b(S_B) \geq b_{\min}$, a cutoff derived from $\mathrm{cov}_{\min}$. Within each band, surprise magnitudes vary by at most $2\times$, so $S_A \cdot S_B$ varies by at most $4\times$. Pairs below the cutoff are provably unable to contribute to promotion.

This is coarse-graining in surprise space: integrating out irrelevant degrees of freedom (low-magnitude pairs that cannot contribute) to focus computation on effective interactions (high-magnitude co-surprises that drive causal discovery). The resulting complexity is $O(E + \sum b_i^2)$, which is near-linear when surprise is distributed across bands.

Deduction: Transitive Closure. Given stable passive models $A \to B$ (lag ℓ_1) and $B \to C$ (lag ℓ_2), the system deduces $A \to C$ with lag $\ell_1 + \ell_2$ and weight $w_{AC} = w_{AB} \cdot w_{BC} \cdot d$, where $d < 1$ is a deductive discount. This discovers relationships at temporal distances beyond the error trace lookback window. In our benchmarks, the link from ignition to ash (lag 3) was never discovered empirically within 80 cycles; deduction found it at cycle 4 from two stable lag-1 links.

Abduction: Hidden Cause Hypotheses. When the system observes effects consistent with a known causal pattern but the cause is not directly observed, abduction hypothesizes that the cause is active. Given observed high values for heat and smoke, and known passive models `fire→heat` and `fire→smoke`, abduction creates a low-precision belief in fire: `(belief fire 0.8 0.3)`.

The low precision is the key. Because precision on the hypothesized cause is low, any observation that would increase it has high information gain. This makes "look for fire" the highest EFE-reducing action—the system spontaneously investigates. We call this the *Sherlock Holmes effect*: the system creates a hypothesis and then seeks evidence, driven entirely by EFE minimization.

Metabolic Selection: The Uniform Arbiter. All three inference types produce atoms subject to the same metabolic regime:

$$E_t = E_{t-1} - c_{\mathrm{rent}} + c_{\mathrm{reward}} \cdot \mathbf{1}[\text{prediction correct}], \tag{5}$$

where $c_{\mathrm{rent}} = 0.02$ per cycle and $c_{\mathrm{reward}} = 0.05$ per correct prediction. An atom that never predicts correctly dies in $\lfloor 1.0/0.02 \rfloor = 50$ cycles.

The metabolic step and reward are implemented directly:

```
; Drain: every atom pays rent each cycle
(= (drain-lifecycle-energy! $type $name)
   (let $current (get-lifecycle-energy $type $name)
     (set-lifecycle-energy! $type $name
       (- $current (get-config metabolic-rate)))))

; Reward: correct predictions earn energy
(= (reward-passive-model! $cause $effect $actual-error)
   (let* (
     ($old-energy (get-pm-energy $cause $effect))
     ($correct (is-prediction-correct? $effect $actual-error))
     ($earned (if $correct (get-config metabolic-boost) 0.0))
     ($new-energy (min (+ $old-energy $earned)
                       (get-config metabolic-energy-cap)))
   ) (sequential
       (remove-atom &state (metabolic-energy passive-model
                            $cause $effect $old-energy))
       (add-atom &state (metabolic-energy passive-model
                         $cause $effect $new-energy))
       done)))
```

This provides principled forgetting (obsolete structure is automatically removed), bounded growth (generation caps combined with metabolic drain), and type-agnostic selection (inductive, deductive, and abductive hypotheses compete on equal footing—only predictive accuracy matters).

3.4 Adaptive Planning

Exhaustive enumeration of action sequences has complexity $O(|A|^d)$. We employ an adaptive beam search whose beam width emerges from the EFE landscape rather than being set as a hyperparameter. The planner uses a single adaptive mechanism—the noise floor—to control all pruning:

$$n(d) = \frac{n_0}{\bar{c} \cdot \gamma^d}, \tag{6}$$

where n_0 is a base noise level, $\bar{c}$ is average model confidence, and $\gamma < 1$ degrades confidence with depth. This produces three behaviors from one mechanism: *adaptive beam width* (flat EFE landscapes produce wide beams; steep ones produce narrow beams); *symmetry breaking* (when one action dominates, the beam collapses); and *emergent horizon* (planning stops when degraded confidence falls below a floor). The beam width is not a hyperparameter—it emerges from the EFE landscape. A hard beam cap narrows with depth via $k_{\max}(d) = \max(2, \lceil k_0 \cdot \bar{c} \cdot \gamma^d \rceil)$. Overall complexity is $O(|A| \times k \times d)$.

3.5 Computational Scaling via Observable Triage

The adaptive planner prunes the *temporal* decision tree; an analogous mechanism prunes the *spatial* observable set each cycle. A computational noise floor $\sigma_{\text{compute}} = \sigma_{\text{base}} / \max(\bar{p}, 0.01)$, derived from average belief precision $\bar{p}$, classifies each observable as *active* (full processing), *dormant* (belief update only), or *frozen* (removed from working memory). Under viability pressure, the floor

tightens by a factor $(1 + 3v_p)$. Structure learning pairs are restricted to observables with recent surprise, which combined with the magnitude-banded pruning of Section 3.3 reduces the effective pair count at 1,000 observables with 5% active from ∼500,000 to ∼130. Four emergent phenomena arise: *attention* (the active set is the system's focus), *habituation* (well-predicted observables are pruned), *vigilance* (the active set expands under threat), and *computational relaxation* (cycles accelerate in predictable environments). Observable triage and structure learning sparsification are implemented; branch-and-bound EFE early termination is designed but not yet implemented.

3.6 Safety Architecture

Safety uses three strata of immutability: *Constitutional* (immutable viability bounds, safety axioms), *Goal* (modifiable via authenticated protocol), and *Learned* (freely modifiable causal links, hypotheses). A post-cycle watchdog hashes constitutional atoms and halts the system on modification. Mutation guards prevent learned-stratum operations from touching higher strata.

3.7 Perception Boundary

For conversational applications, a language model translates natural language input into structured observations: utterance type, topic list, and scalar estimates of partner comprehension, rapport, and urgency. Each enters the cognitive core as a precision-weighted tuple (o, v, p, t)—identical in schema to a hardware sensor reading. Because the cognitive core operates exclusively on this uniform schema, the language model is an interchangeable perception adapter: any model capable of structured extraction may be substituted, and applications with non-linguistic input (sensor arrays, APIs, robotic platforms) bypass it entirely. Language model parses enter at low precision (0.25–0.35); hardware sensors at high precision (0.80–0.95). When the cognitive core selects `wait`, no generation is triggered—the system remains silent rather than producing fluent but contentless output (the *epistemic firewall*). On the outbound path, the core packages a structured intent; the language model performs sentence construction within these constraints. A proprioception module re-parses the system's own output and injects the discrepancy as a verbalization-fidelity observation.

4 Evaluation

4.1 Validation Methodology

Validation proceeds at three tiers. *Tier 1*: 22 MeTTa modules define the canonical architecture. *Tier 2*: Python executable specifications encode identical logic, thresholds, and formulas, differing only in host language—these validate the dynamics. *Tier 3*: a pure Python MeTTa evaluator (`dagaz_runtime.py`, ∼750 lines) loads and executes the canonical MeTTa source through generic pattern

Table 1. EFE action selection results. All pass without thresholds or mode switches.

Scenario	Condition	Selected	Pass
Nominal	All near preferences	wait	✓
High threat	threat = 0.7	retreat	✓
Uncertain terrain	precision = 0.3	observe	✓
Rough terrain	terrain = 0.7	retreat	✓
Low power	power = 0.25	wait	✓

matching, implementing no domain-specific logic. A stateless orchestrator connects this to a language model (Llama 3.2 3B via Ollama in our tests) for end-to-end conversational operation; any model capable of structured extraction may be substituted, and non-conversational applications require no language model. Results below are from Tier 2 unless noted.

4.2 Core Dynamics: Action Selection and Reasoning

We first validate that the two central mechanisms—EFE-driven action selection and the Peircean reasoning triad—produce correct behavior in isolation before testing their interaction in the integrated reef scenario (Section 4.5).

Five action selection scenarios with three actions across three observables confirm that behavior emerges from EFE minimization without thresholds or mode switches (Table 1). In the high-threat scenario, the viability adjustment provides retreat with a bonus of -0.22, transforming it from the highest-cost action to the lowest-EFE action. Scaling to 50 actions and 24 observables confirms linear EFE computation: 0.04 ms at 3 actions to 0.67 ms at 50 (16.4×, vs. theoretical 16.7×).

Seven reasoning scenarios exercise the full Peircean triad under metabolic selection (Table 2). The most significant result is Scenario 4, the Sherlock Holmes effect: abduction creates a low-precision belief in a hidden cause; because precision is low, information gain for confirming observations is high; EFE minimization then selects the investigative action. No module was programmed to produce investigation behavior—it emerges from the interaction of abduction, the precision–information-gain relationship, and EFE minimization. Scenario 6 validates the uniform metabolic arbiter: incorrect hypotheses of all three types die within ∼30 cycles.

4.3 Adaptive Planning

The adaptive beam search is tested across eight scenarios at 3 actions and four additional scenarios scaling to 50 actions (Table 3).

At 50 actions, the planner evaluates 195,300 of 15.6 billion exhaustive paths—an 80,000× reduction—because steeper EFE landscapes break symmetry earlier. In the flat landscape scenario, the beam widens to nearly exhaustive search: the

Table 2. Unified reasoning triad results. All seven scenarios pass.

#	Scenario	Pass
1	Induction: 4-node causal chain discovered within 3 cycles	✓
2	Deduction: transitive link (lag 3) found at cycle 4; never empirically	✓
3	Abduction: hidden cause hypothesized from observed effects	✓
4	Sherlock Holmes: abduction drives spontaneous investigation	✓
5	Full triad: induction → deduction → abduction in sequence	✓
6	Metabolic death: wrong hypotheses die within ∼30 cycles	✓
7	Competing causes: asymmetric evidence disambiguates	✓

Table 3. Adaptive planning results. Pruning improves with action count: steeper EFE landscapes break symmetry earlier.

Scenario	Exhaustive	Adaptive	Correct?
Depth 7, beam 2 (3 actions)	2,187	42	✓
Depth 7, early symmetry break	2,187	18	✓
Flat landscape (deliberation)	27	24	✓
Confidence degradation cutoff	2,187	15	✓
10 actions, depth 6	1,000,000	7,196	✓
20 actions, depth 6	64,000,000	2,172	✓
50 actions, depth 6	15.6 B	195,300	✓

system correctly recognizes that alternatives are indistinguishable. With tightened parameters, the same planner achieves 1.4–3.0× additional speedup with 5/5 answer preservation across random seeds.

4.4 Scaling and Sensitivity

Two analyses validate that the architecture's core mechanisms remain sound at scale.

Metabolic Parameter Sensitivity. We swept 110 parameter pairs (11 rent × 10 reward values) across an induction–selection scenario (Table 4). The key finding is that the metabolic economy depends primarily on a single dimensionless ratio $c_{\text{reward}}/c_{\text{rent}}$. Near-universal health obtains above ratio 1.0, with near-universal failure below 0.5. The dominant failure mode is starvation (correct hypotheses die), which is the safer direction. Induction itself never fails—the Hebbian mechanism is parameter-independent. Scaling validation at 6–24 observables confirms 78–89% healthy parameter space, consistent with the 67% at 3 observables.

Hebbian Accumulator Scaling. To validate the banded pruning optimization of Section 3.3, we measured pair reduction and causal discovery fidelity across

Table 4. Metabolic sensitivity analysis. The healthy basin occupies 67% of parameter space, governed by the dimensionless ratio $c_{\text{reward}}/c_{\text{rent}}$.

Metric	Value
Healthy configurations	74/110 (67%)
Failures from starvation	33/36 (92%)
Failures from bloat	3/36 (8%)
Transition ratio ($c_{\text{reward}}/c_{\text{rent}}$)	≈ 1.0
Induction failures	0/110

Table 5. Banded pruning pair reduction vs. brute force. Significant misses counts pairs whose single-cycle covariance exceeds the contribution threshold but were pruned. Zero at all scales.

Observables	Brute Pairs	Pruned Pairs	Reduction	Sig. Misses
14	552	456	1.2×	0
50	5,674	2,345	2.4×	0
200	60,829	9,851	6.2×	0
500	312,764	17,194	18.2×	0
1,000	1,840,778	94,818	19.4×	0

five scales from 14 to 1,000 observables using synthetic causal chain scenarios (3 chains of length 4, 30 cycles).

Three findings emerge (Table 5). First, pair reduction grows with scale: from negligible at 14 observables to 19.4× at 1,000, because larger observable spaces produce more low-surprise noise traces landing in pruned bands. Second, significant false negatives are zero at every scale—the $b_{\min}$ cutoff, derived from metabolic parameters via Eq. (4), correctly identifies the boundary between productive and unproductive pairs. Third, causal discovery is preserved exactly: banded pruning discovers the same passive models as brute force at all tested scales. Regime sensitivity analysis at 200 observables confirms the optimization is most effective when most needed: in high-noise conditions, pair reduction reaches 35.9× with zero significant false negatives, because noise traces land in low-magnitude bands that are pruned entirely.

At 1,000 observables, 93.6% of traces fall in bands 0–3, all pruned when paired with each other ($0 + 3 < b_{\min} = 7$). Only 4.5% of traces in bands 5–7 participate in most pairings, reducing the effective population from 1,920 to ∼86. Structure learning additionally maintains bounded graph density: at 24 observables (276 possible pairs), only ∼11% become links, compared with 42% at 4 observables. The metabolic economy prunes spurious links natively without explicit regularization.

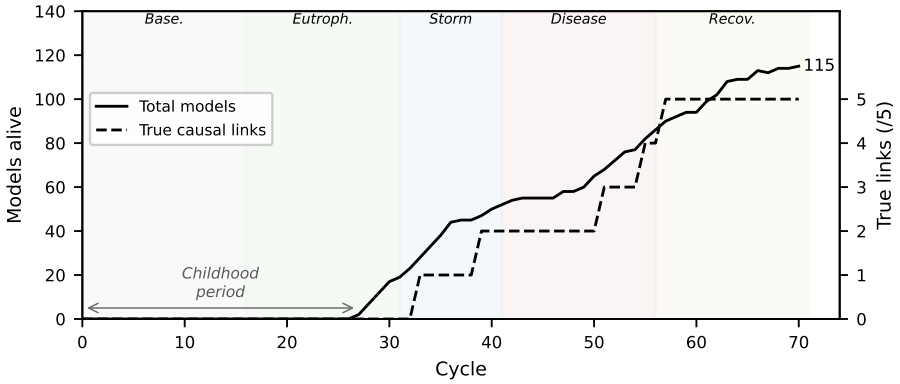

Fig. 2. Metabolic lifecycle across 70 cycles (boundary fix engine). The childhood period (cycles 0–27) reflects near-zero initial precision suppressing premature model creation. All 5/5 true causal links discovered; only 3 incorrect models pruned.

4.5 Reef Monitoring Scenario

To test the architecture in an integrated setting, we constructed a coral reef monitoring scenario: 14 environmental and platform variables across 70 cycles spanning five ecological phases (baseline, eutrophication, storm, disease, recovery). The environment contains four hidden causal chains with temporal lags, including a pathogen that is never directly observable.

Causal Discovery. All five true causal links are discovered and survive to cycle 70: the complete eutrophication chain (nutrient-load $\rightarrow$ algae-coverage, nutrient-load $\dashv$ dissolved-oxygen, dissolved-oxygen $\rightarrow$ coral-health, coral-health $\rightarrow$ fish-activity) and the storm mechanism (current-strength $\rightarrow$ turbidity). Hub detection identifies nutrient-load as a degree-4 node. The emergent childhood period (cycles 0–27, Fig. 2) confirms that near-zero initial precision suppresses premature hypothesis generation. Only 3 incorrect models are pruned by cycle 70, compared with 48 under the baseline engine.

Ablation Study. With EFE-driven action selection, we swept info-gain weight and viability weight across 90 configurations (without abduction) and 80 configurations (with creative abduction). Three emergent behaviors were tracked: curiosity, investigation, and the Sherlock Holmes effect.

The ablation (Table 6) reveals that EFE alone is insufficient for the Sherlock Holmes effect—creative abduction is a necessary component. Without it, the system cannot investigate what it does not know it does not know, regardless of parameter tuning (0/90 across all configurations). Configurations producing investigation sacrifice exploration and vice versa—resolving this mode transition is the primary integration challenge.

Table 6. Ablation results. Creative abduction is necessary for investigation—not a convenience.

Emergent behavior	Without abduction	With abduction
Curiosity	42/90 (47%)	0/80 (0%)
Investigation	0/90 (0%)	80/80 (100%)
Sherlock Holmes	0/90 (0%)	80/80 (100%)
≥3/5 true causal links	42/90 (47%)	20/80 (25%)

Table 7. Integration results across 20 parameter configurations. The boundary condition fix achieves 100% gold standard (Sherlock Holmes + ≥3 causal links)—the only variant to do so.

Metric	Baseline	Gates	Boundary Fix
Curiosity + Investigation	0%	100%	100%
Sherlock Holmes effect	100%†	40%	100%
Avg true causal links (/5)	2.0	2.6	4.0
Avg metabolic deaths	48	44	7
Gold standard	0%	0%	100%
First creative abduction	cycle 3	cycle 8	cycle 38
Design-principle compliant	✓	∼	✓

†Baseline's 100% Sherlock rate is misleading: hypotheses created from noise, not anomaly detection.

Resolving the Integration Challenge. Diagnosis revealed that uniform initial priors (value 0.5, precision 0.5) cause the system to interpret boot-up calibration errors as evidence of hidden causes, generating latent hypotheses for nearly every observable. An initial solution using epistemic gates (observation coverage ≥60%, belief stability < 0.15) was effective but violated the Bottom-Up and Emergent design principles.

The resolution requires no new mechanisms—only correct boundary conditions:

Near-zero initial precision. Beliefs initialize at $p_0 = 0.01$. The surprise formula $S = \frac{1}{2}p \cdot e^2$ is linear in precision: at $p = 0.01$, a prediction error of $e = 0.4$ yields $S = 0.0008$—below the cognitive threshold by two orders of magnitude. An observation-driven precision ratchet integrates p upward by $\Delta p = 0.03$ per observation, creating an emergent childhood period of ∼15 observations.

Metabolic capital for abduction. Creative abduction costs capital ($c_{\text{abd}} = 0.5$), accumulated from successful predictions (+0.02 each). This naturally sequences the Peircean triad: induction (free) precedes abduction (expensive). Epistemic shielding dampens the viability penalty when a latent hypothesis exists for the threatened observable, permitting investigation under threat.

The boundary fix (Table 7) achieves 100% gold standard across all 20 configurations. The dramatically lower death count (7 vs. 44–48) confirms hypotheses are created at the right time. The extended calibration period (first abduction

at cycle ∼38) means the agent has built a solid inductive world model before attempting abduction. This progression—from threshold-based patches to correct initial conditions—illustrates a general principle: when emergent behavior fails, diagnose the initial conditions before patching the dynamics.

5 Discussion

5.1 Limitations

Runtime status. The MeTTa source is canonical, written against the MeTTa language as described by Goertzel et al. [19]. Two bugs in Hyperon 0.2.10 affect native execution: a cons-cell pattern matching failure breaking recursive list functions, and a trie index corruption after ∼1,000 accumulated atoms. An architectural three-space separation (static code in `&self`, dynamic state in `&state`, immutable ontology in `&ontology`) prevents the trie bug from triggering. The Python MeTTa evaluator supports both sources of MeTTa nondeterminism: function dispatch explores multiple definitions for a given pattern, and `collapse(match ...)` iterates all bindings from a space query. Bare `match` currently returns only the first binding—a bounded limitation that does not affect the cognitive cycle, where multi-result queries use `collapse`.

Scale. Benchmarks validate 3–50 actions, 4–24 observables, and Hebbian scaling to 1,000 observables. Two open questions remain: passive model count growth at 500+ observables (whether metabolic pruning keeps pace with hypothesis generation), and the interaction between banded pruning and the adaptive metabolic bootstrap (if the system adjusts its own rent/reward rates, $b_{\min}$ must be rederived).

Integration dynamics. The boundary condition fix introduces two new parameters (precision ratchet rate, abduction capital cost) whose sensitivity across domains has not yet been characterized. However, the mechanism operates through the existing surprise equation $S = \frac{1}{2}p \cdot e^2$ rather than external thresholds, so any reasonable near-zero initial precision will produce suppression during boot-up. The evaluation uses a single scenario; generalization to other domains has not been verified.

5.2 Related Work

Active Inference implementations using deep neural networks [3,15] achieve scalability at the cost of transparency. Champion et al. [4] implement discrete decision-making agents with explicit state spaces. The toolkits pymdp [11] and RxInfer.jl [2] focus on efficient inference within a given generative model; our work addresses the orthogonal problem of *discovering and maintaining* the generative model via induction, deduction, and abduction under metabolic selection. These approaches are complementary—a mature inference backend could serve within a single cycle of our architecture.

Among cognitive architectures, SOAR [13] uses production systems with chunking, ACT-R [1] models cognition through Bayesian activation, and LIDA [5]

implements global workspace theory. These achieve breadth through engineered module interactions. Our architecture achieves breadth through a single optimization principle applied uniformly across all subsystems. The metabolic selection mechanism has no direct analogue—SOAR retains all chunks indefinitely; ACT-R's base-level decay operates on frequency rather than predictive accuracy.

The metabolic economy shares structural similarities with OpenCog's Attention Economy [8], but uses simpler fixed-rate dynamics integrated with the Peircean triad. The adaptive planner relates to hierarchical planning with adaptive abstraction [20], differing in that beam width emerges from the EFE landscape. The use of abduction to drive investigation has been explored [12], but our contribution is showing that this behavior emerges from EFE minimization without a dedicated investigation module. The neural-symbolic boundary differs from hybrid approaches [7,14] in positioning neural models exclusively at the perceptual boundary, governed by the same precision-weighted Bayesian update as all other observations.

6 Conclusion

We have presented a cognitive architecture in which a single optimization principle—minimize Expected Free Energy—produces unified perception, action selection, causal reasoning, temporal planning, and honest self-knowledge. Three forms of inference operate as hypothesis generators under uniform metabolic selection. An adaptive planner achieves up to 80,000× reduction in evaluations at 50 actions. A magnitude-banded covariance pruning scheme reduces pairwise Hebbian scaling to near-linear with zero significant false negatives at 1,000 observables, with its threshold derived from the metabolic parameters governing hypothesis selection.

The most notable result is the Sherlock Holmes effect: the spontaneous emergence of investigation from the interaction of abductive inference, epistemic value, and EFE minimization. An ablation study confirms that creative abduction is necessary—across 90 configurations without abduction, investigation never emerges. The integration challenge—distinguishing boot-up ignorance from genuine anomaly—was resolved not by adding mechanisms but by correcting boundary conditions: near-zero initial precision and metabolic capital for abduction produce an emergent childhood period with no gates or thresholds, achieving 100% Sherlock Holmes effect across all parameter configurations tested.

Characterizing the precision ratchet and capital cost across additional domains, closing the meta-parameter bootstrap loop, and completing native MeTTa execution on Hyperon are the focus of ongoing work.

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